9-3 Various Information Systems

Point!

1 Information Systems

(1) ([1]**TPS- Transaction Processing System**) : A system that records and processes the company's routine daily transactions

<Example> Point of Sales (POS) System, Airline Reservation System

(2) ([2]**MIS - Management Information System**) : A system that provides periodic reports to manageres to assist them in decision-making.

<Example> Monthly Sales Reports, Inventory Reports.

(3) ([3]**DSS - Decision Support System**): A system that supports manageres in making non-routine decisions through modeling and simulaions.

<Example> "What if" Analysis, Business scenario planning

(4) ([4]**EIS - Executive Information System**): A system that provides summarised information to executive level about the overall performance of the organization.

<Example> Executive Dashboards, KPIs

2 System Integration

(1) ([5]**Vertical Integration**): Linking systems at different levels of the organization (such as TPS with MIS).

(2) ([6]**Horizontal Integration**): Linking the systems at the same level across different departments (such as the sales system with the inventory system).

3 Enterprise Resource Planning (ERP)

(1) ([7]**Enterprise Resource Planning System**) : An integrated system that combines various business functions such as accounting, human resources, supply chain, and customer relationship mnagaement into a single system

Warm Up

Answer the following questions.

(1) Choose the term that best fits into the blanks [1] to [3] in the following sentences from the options **A** to **E** below, and answer using the letters.

> Our daily lives are supported by ([1]) composed of humans, computers, and other information and communication technology equipment. Examples include ([2]) where data is collected and managed at the time a product is sold, and ([3]) where data from satellites is obtained to determine your present location and the operation status of buses and other transportation systems.

A POS system **B** Information systems **C** GPS (Global Positioning System)
D ITS (Intelligent Transport System) **E** SNS (Social Networking Service)

Explanation

(1) [1] **B** [2] **A** [3] **C**

Try

1 Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [4] with the appropriate terms.

> A system that utilizes a network to collect, share, and transmit various data and information is referred to as an (　[1]　). Examples of ([1]) include (　[2]　), which is a system that transmits and receives road information to resolve issues such as traffic accidents and congestion. In addition, a (　[3]　) is a system used in convenience stores and supermarkets to collect and manage data at the point of sales.

(2) What is the term for the satellite positioning system that collects location information for any place on Earth?

Exercise

1 Choose the most appropriate explanation of a POS system from the options **A** to **D** below, and answer using the letter.

A A system that uses artificial satellites to determine current locations.
B A system that operates by utilizing networks to collect, share, and transmit various data and information.
C A system used in convenience stores and other shops to manage product sales information.
D A system for transmitting and receiving road information to resolve issues such as traffic accidents and congestion.

2 Choose the most appropriate explanation for vertical integration from the options **A** to **D** below, and answer using the letter.

A Linking systems at the same level across different departments of the organization..
B Linking systems at different levels of the organization.
C Linking systems at a single level within a specific department.
D Linking systems between two unrelated departments